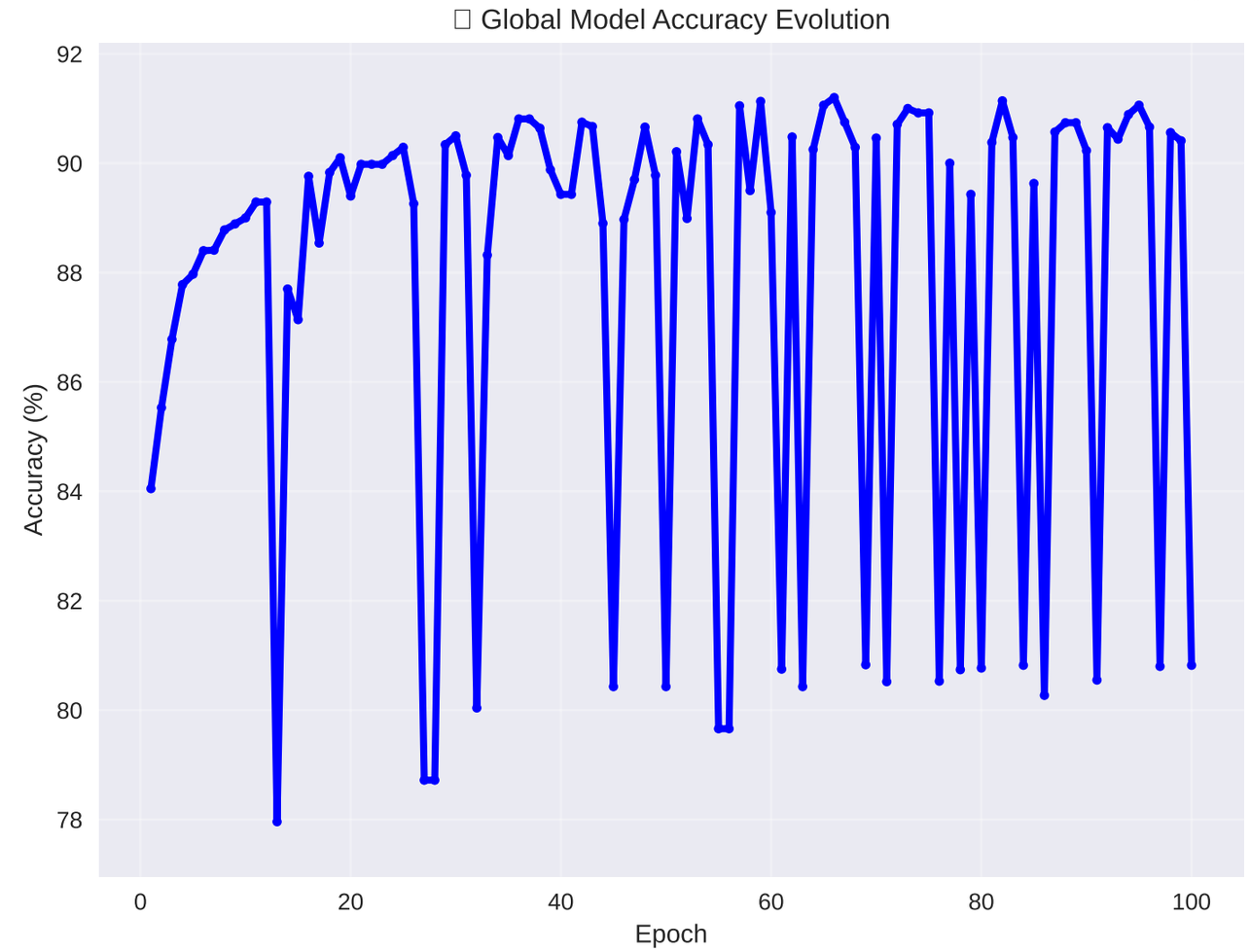
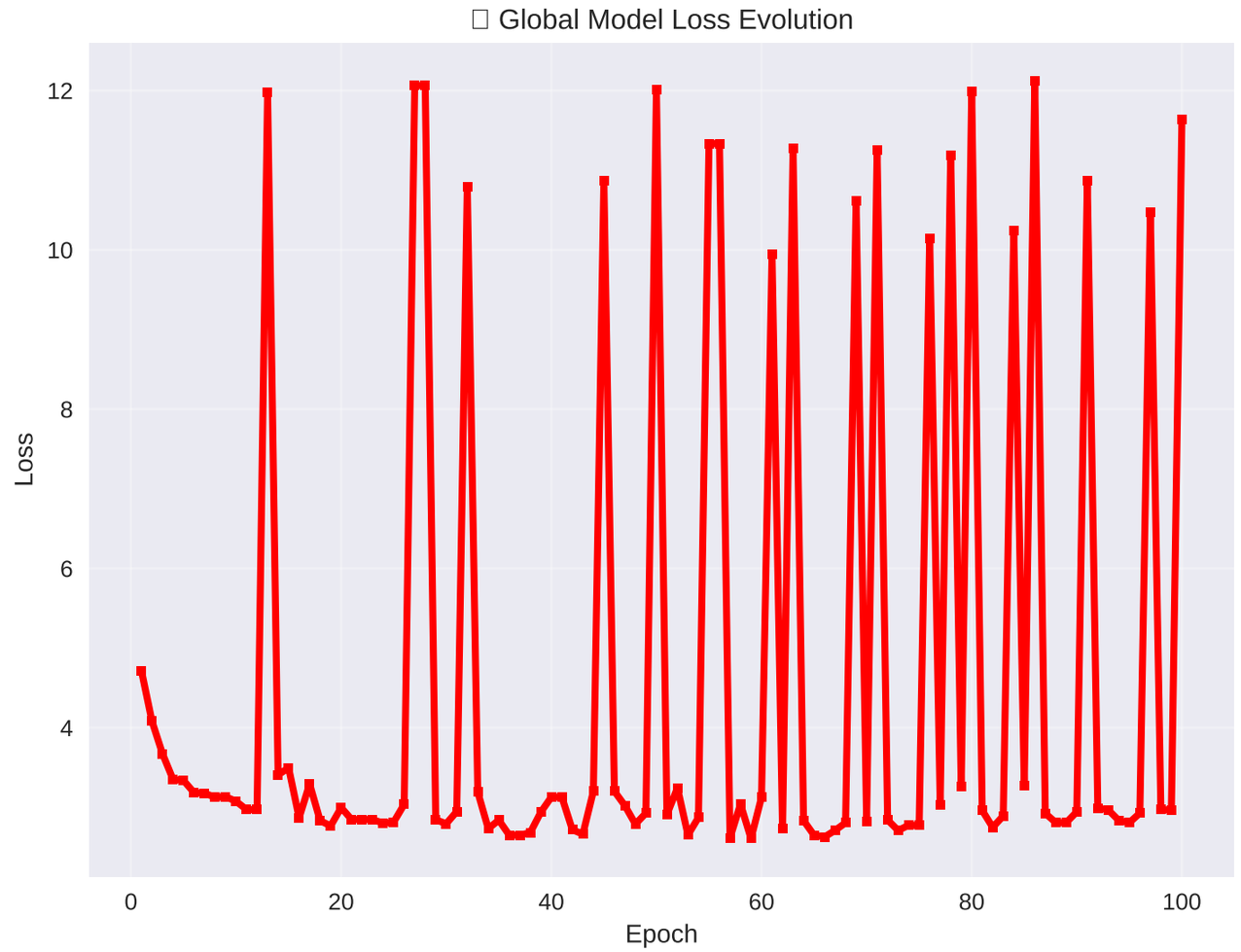
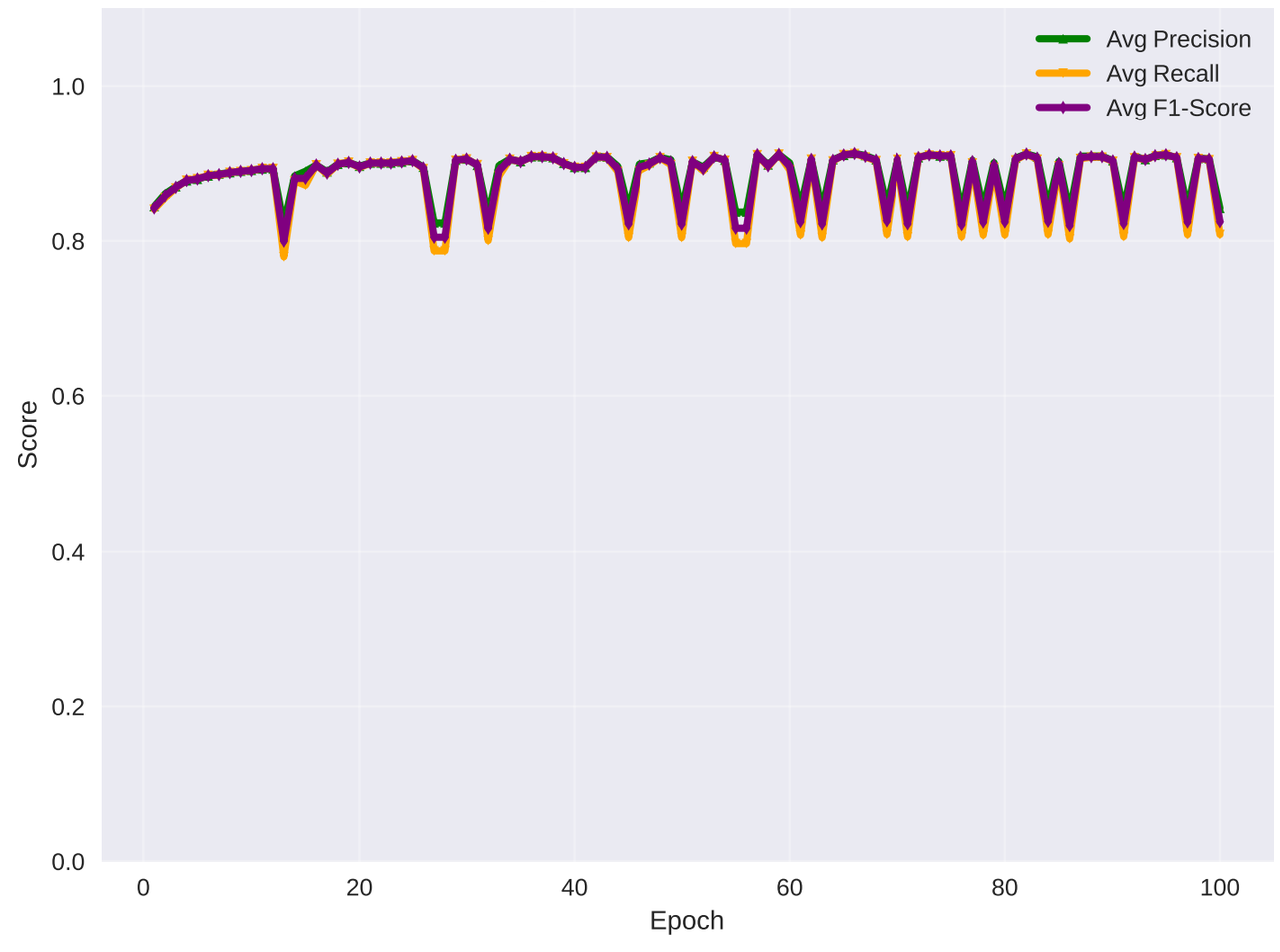
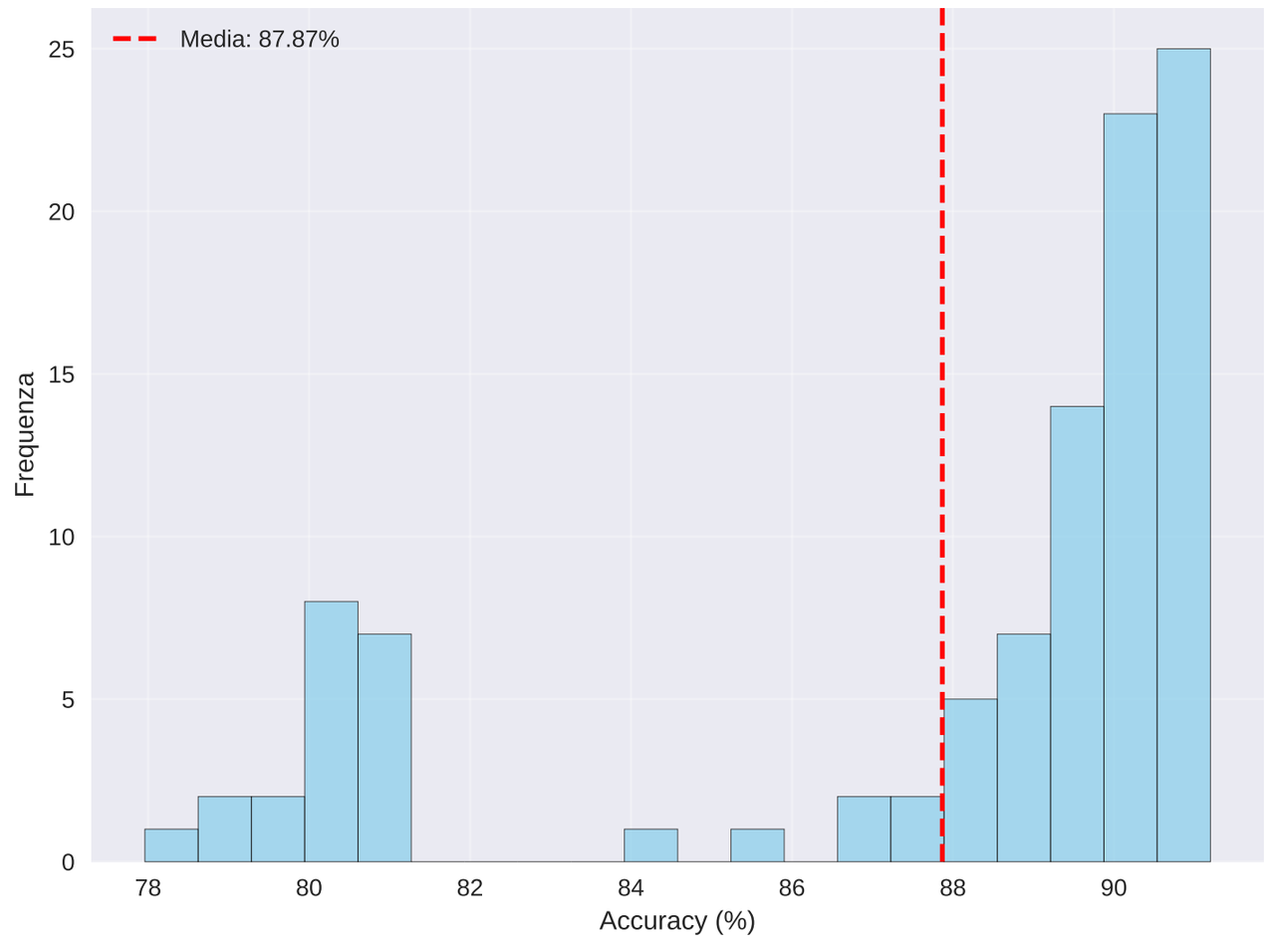
Federated Learning - Main Metrics Overview

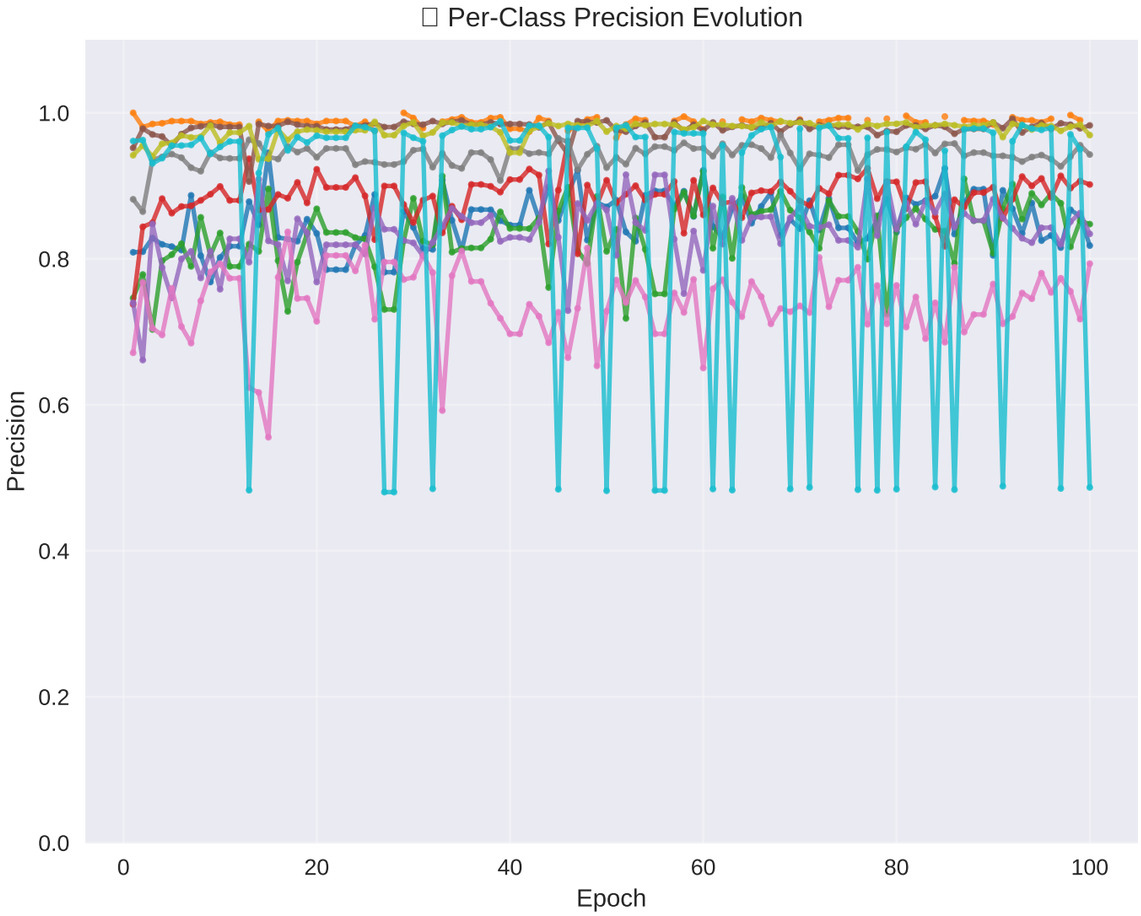


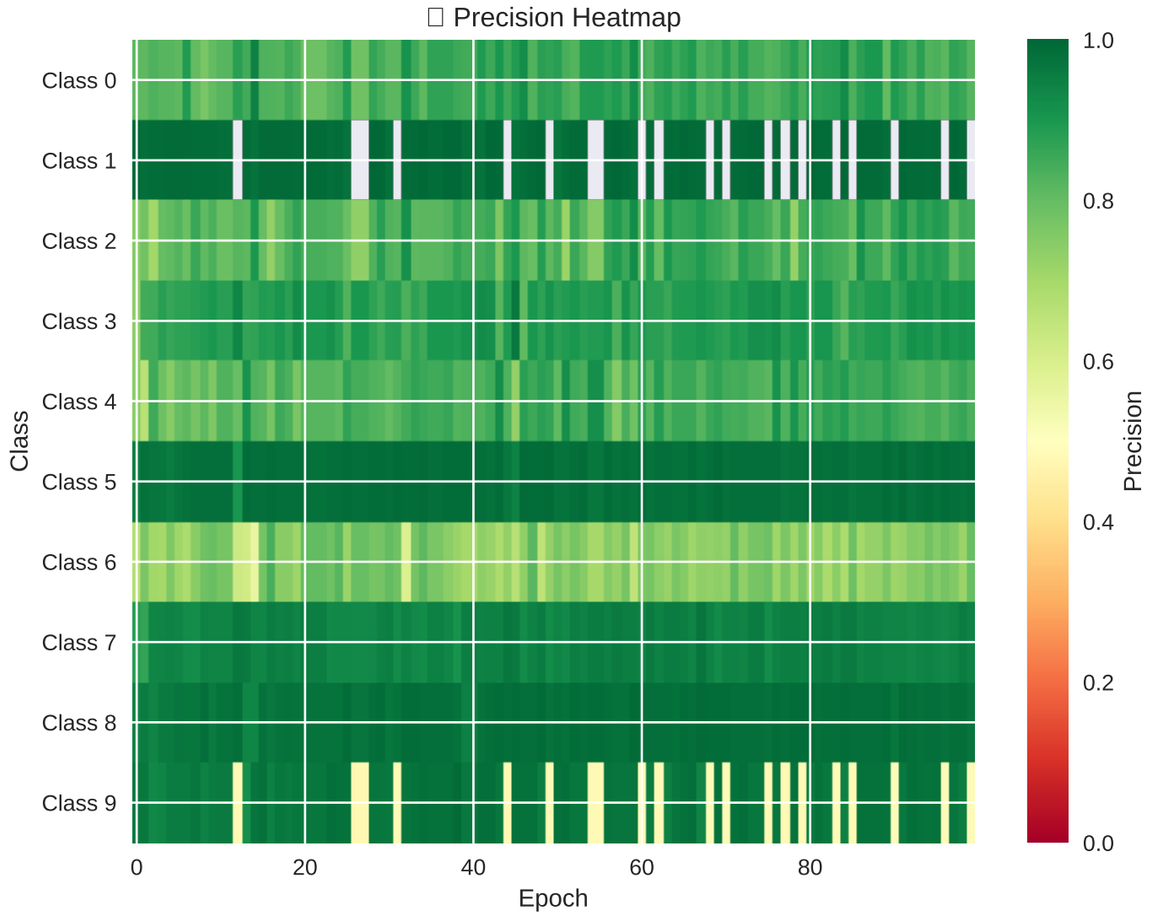


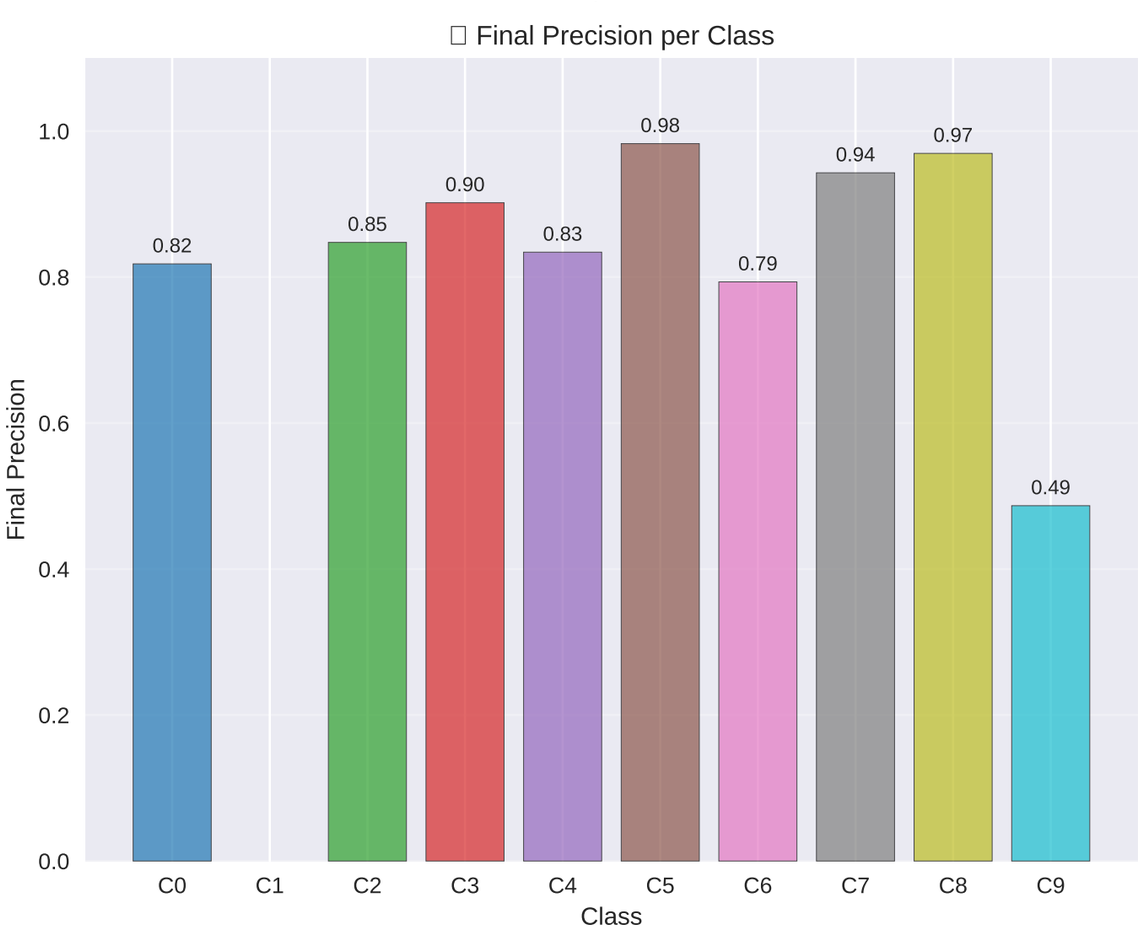


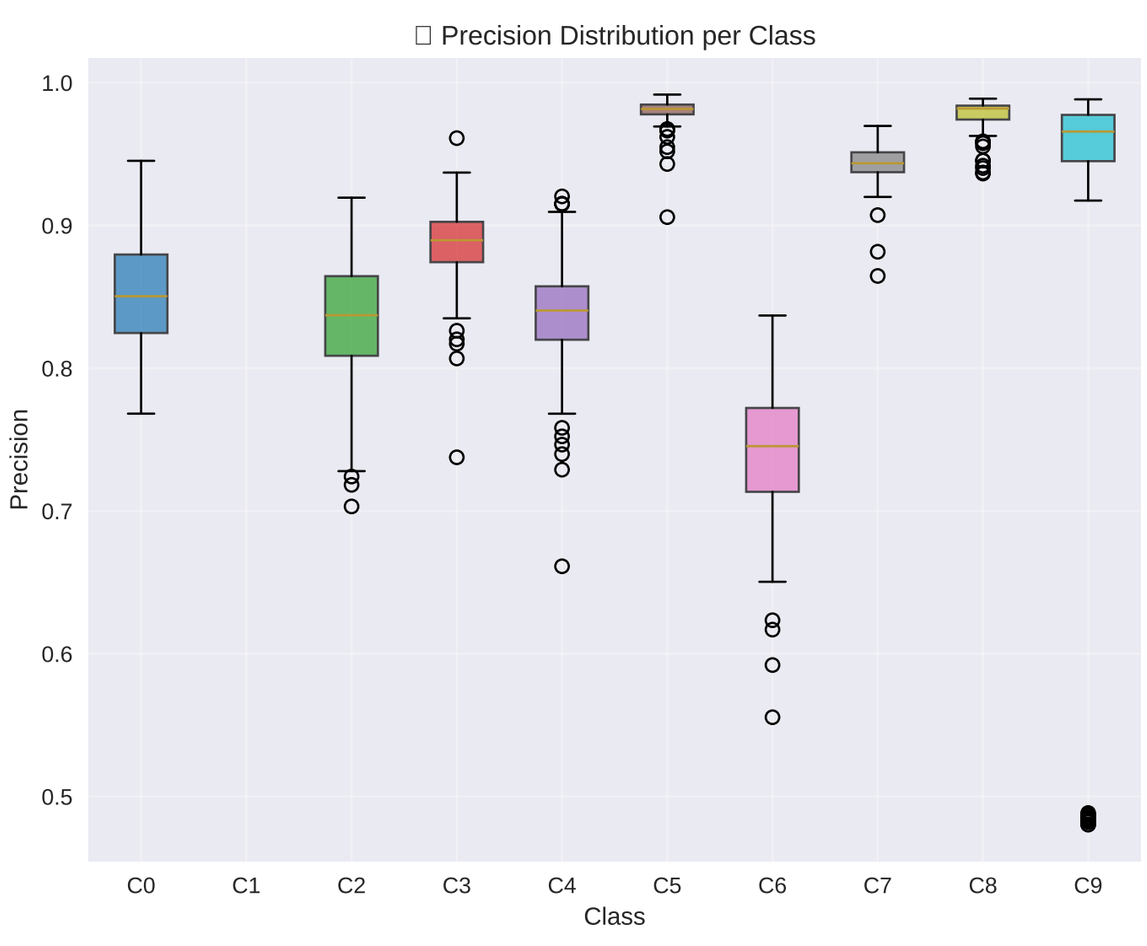


Per-Class Precision Analysis



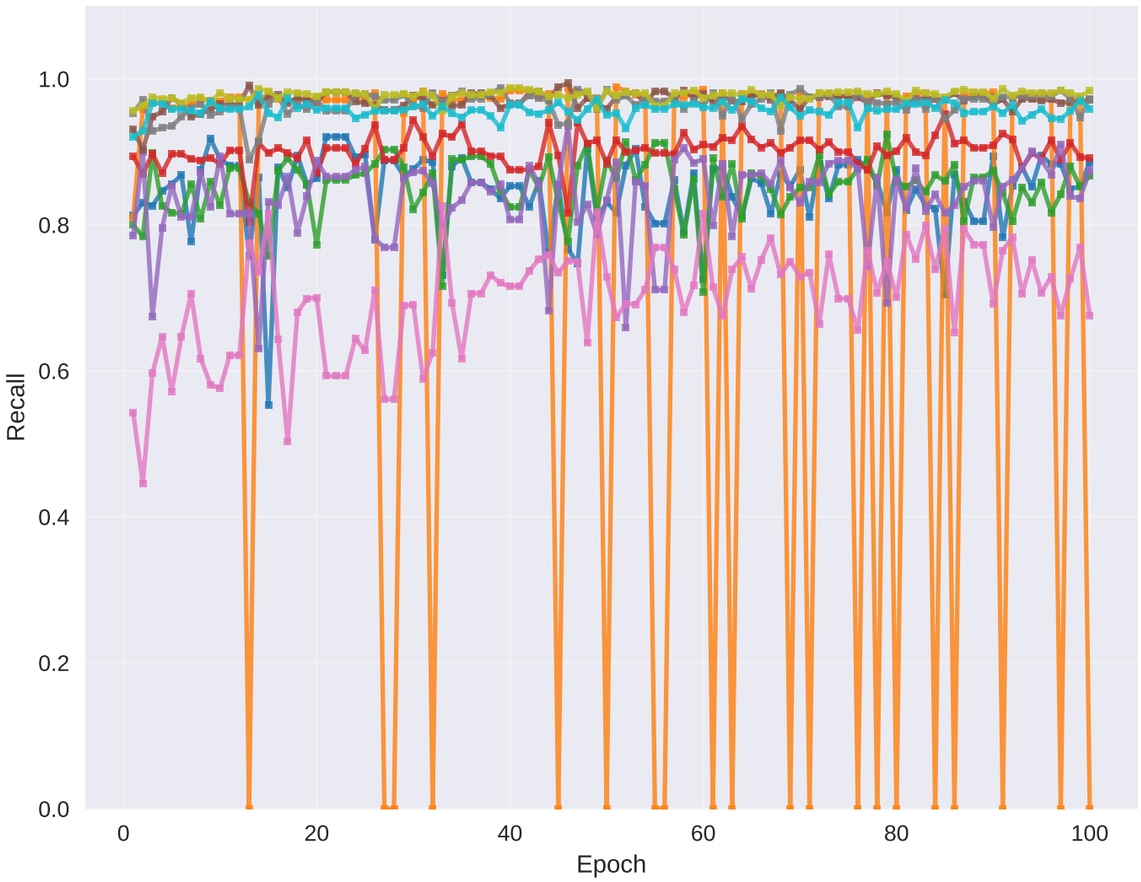




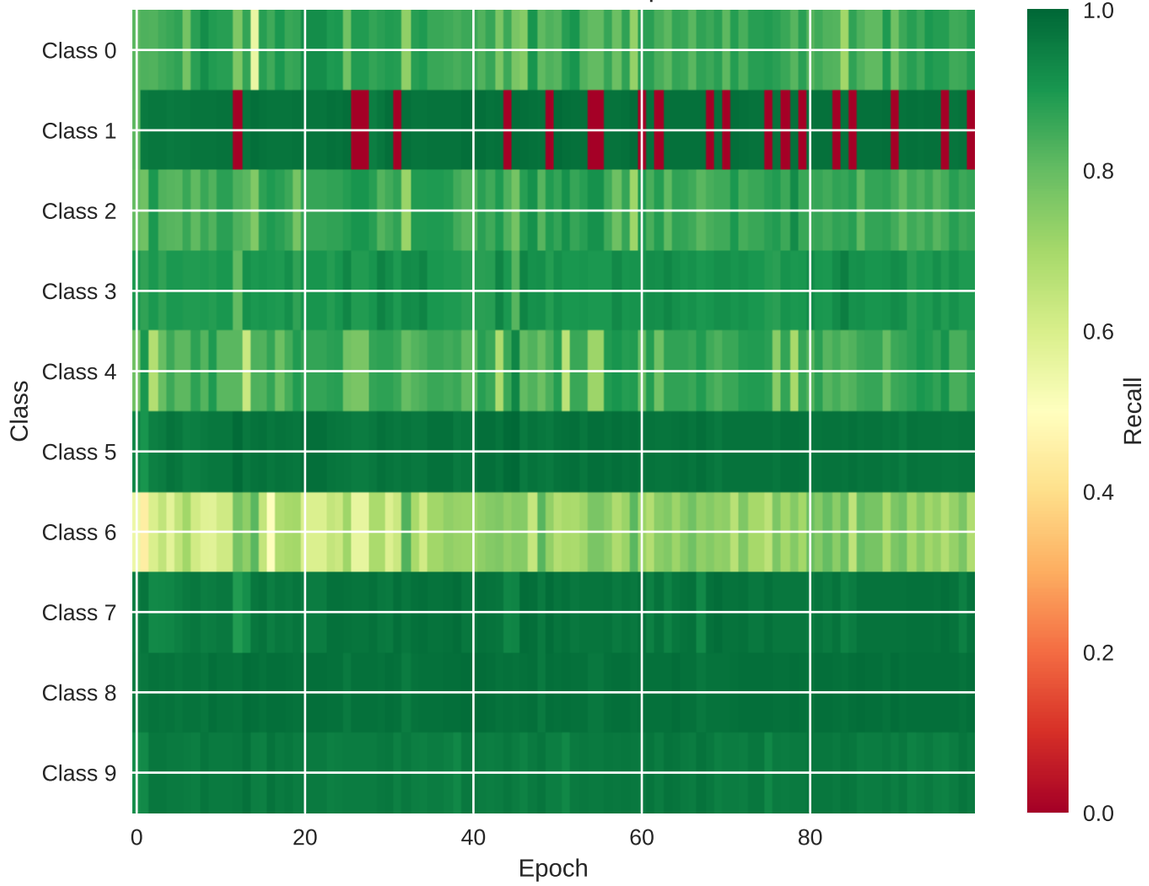


Per-Class Recall Analysis

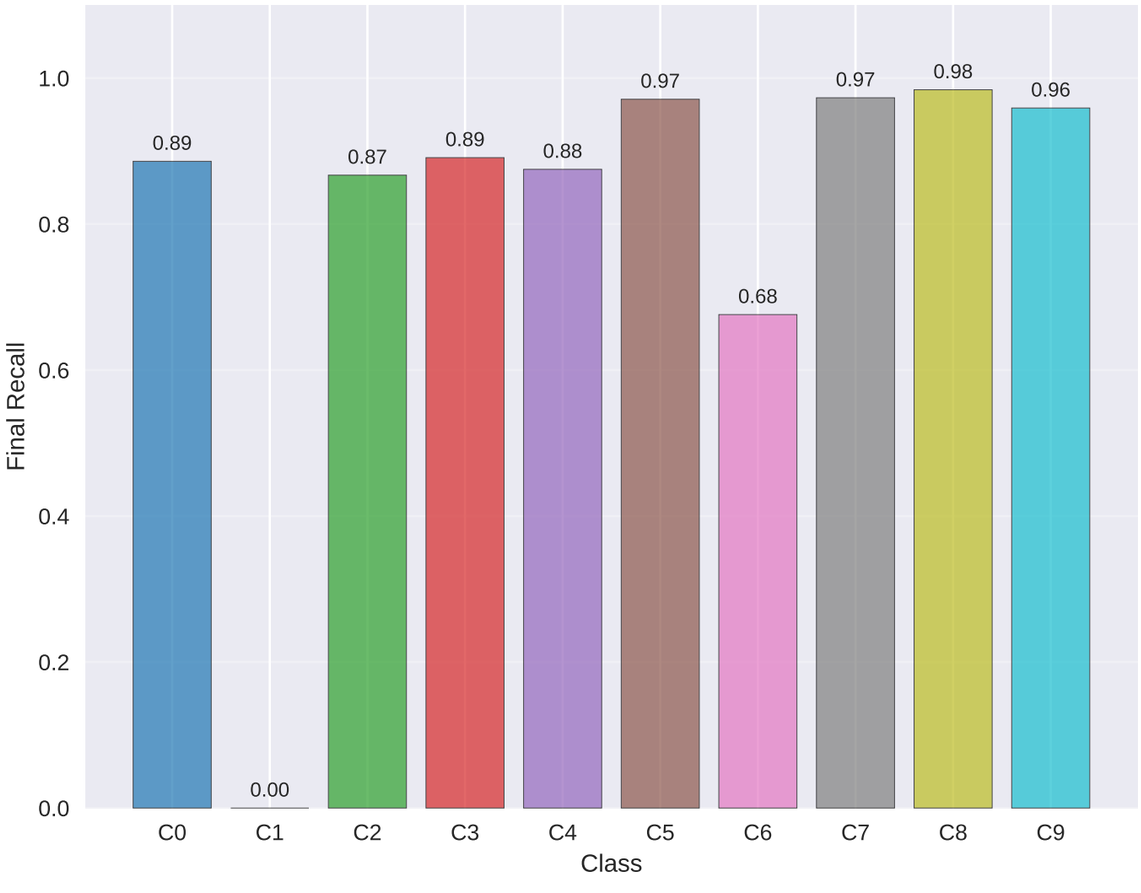
 Per-Class Recall Evolution



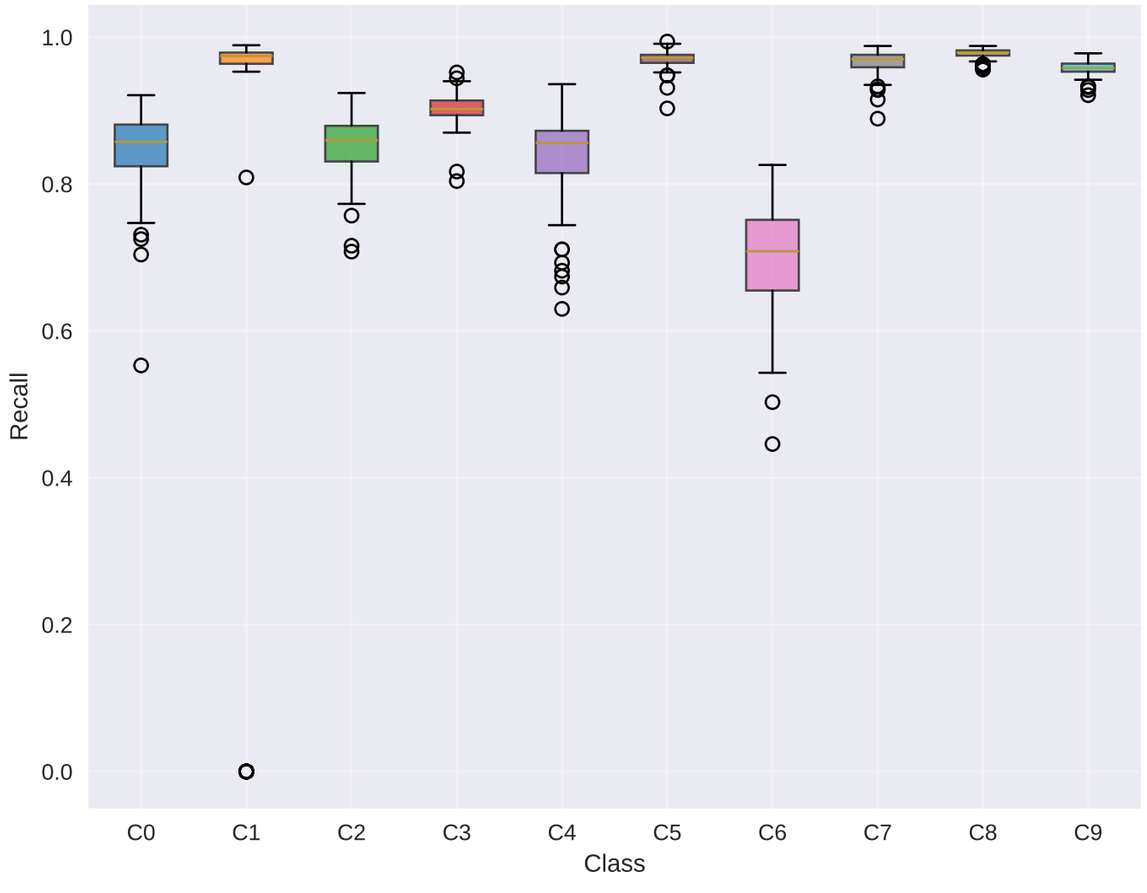
 Recall Heatmap



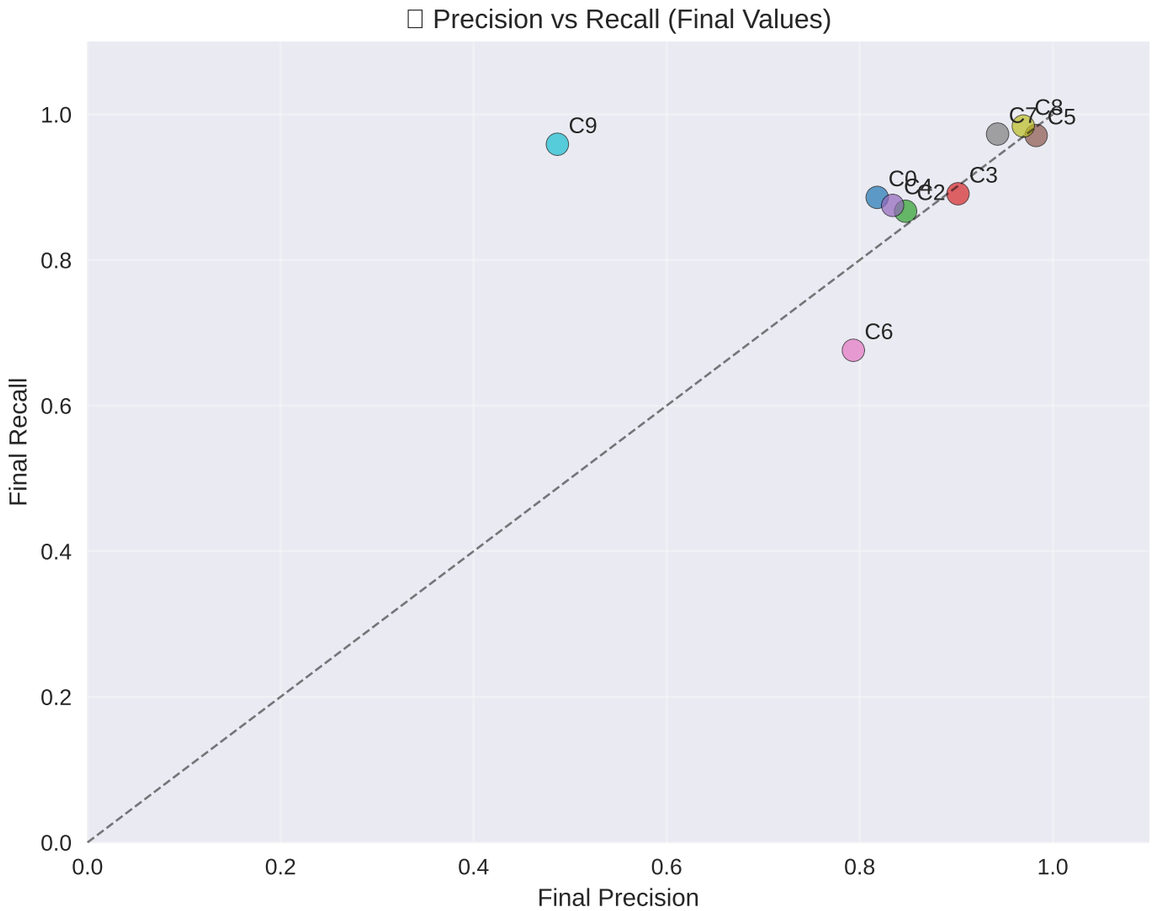
 Final Recall per Class

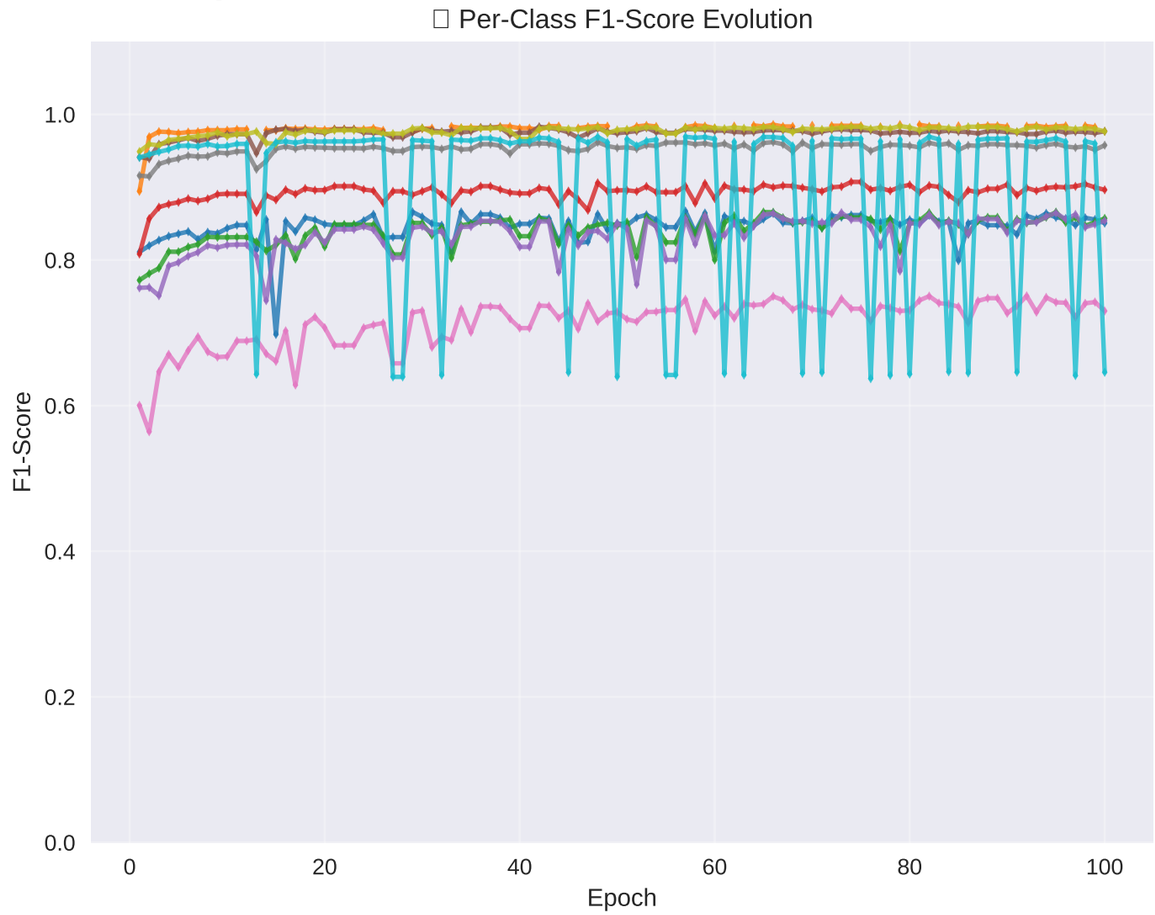


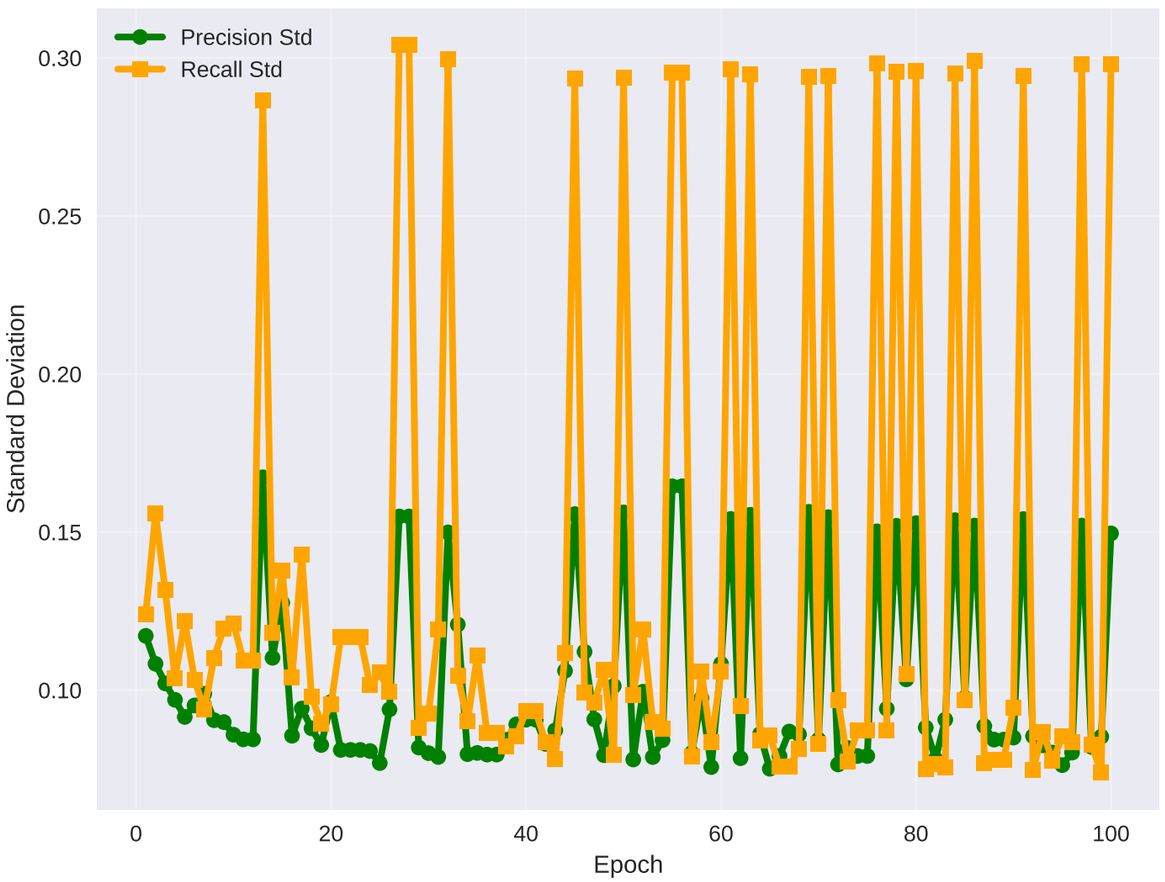
 Recall Distribution per Class

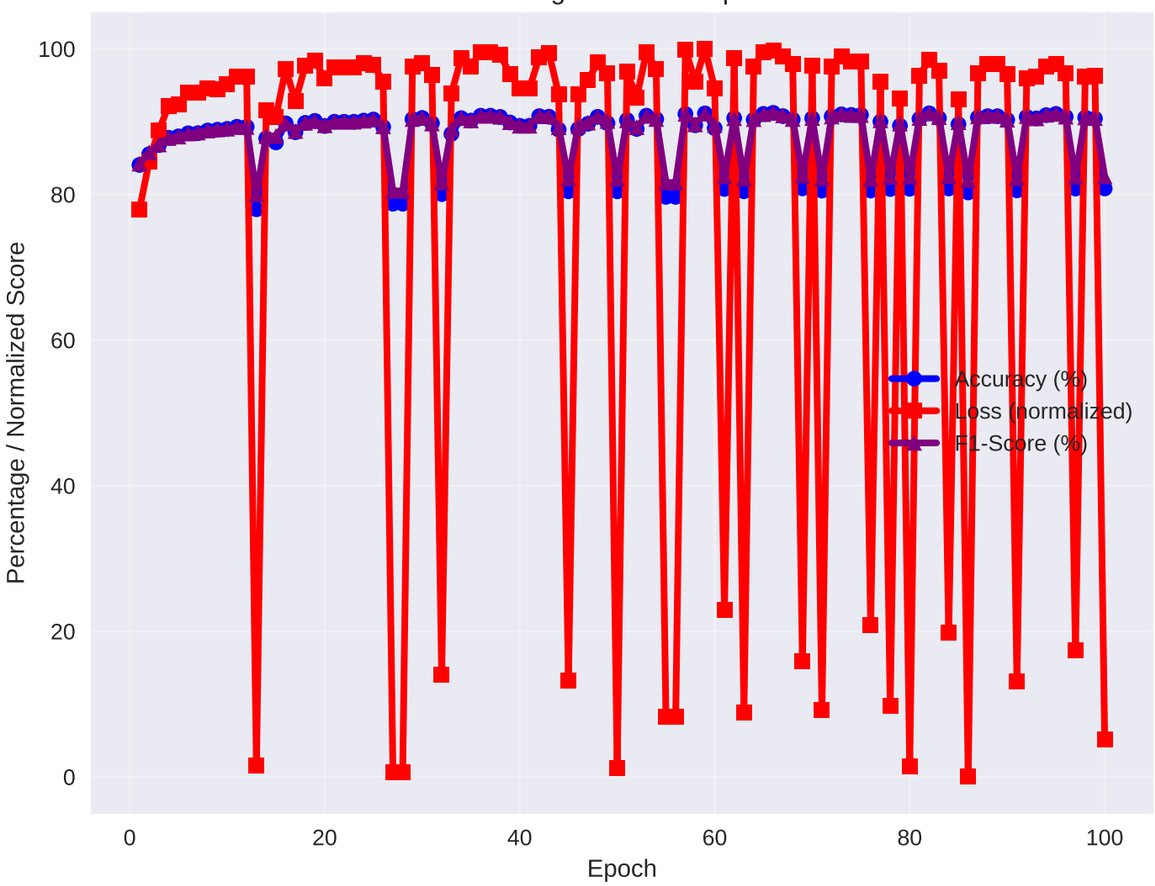


Comparative Analysis









Performance Summary Report

```
FEDERATED LEARNING - PERFORMANCE SUMMARY
========================================

FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-09-05 20:05:32
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 80.82%
• Loss: 11.6343
• Average Precision: 0.842
• Average Recall: 0.808
• Average F1-Score: 0.825

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: -3.23%
• Riduzione Loss: -6.9214
• Stabilità Accuracy (CV): 0.0460

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (nan)
• Peggiore Precision: Classe 1 (nan)
• Migliore Recall: Classe 8 (0.984)
• Peggiore Recall: Classe 1 (0.000)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.818, R=0.886, F1=0.851
• Classe 1: P=nan, R=0.000, F1=nan
• Classe 2: P=0.848, R=0.867, F1=0.857
• Classe 3: P=0.902, R=0.891, F1=0.896
• Classe 4: P=0.834, R=0.875, F1=0.854
• Classe 5: P=0.983, R=0.971, F1=0.977
• Classe 6: P=0.793, R=0.676, F1=0.730
• Classe 7: P=0.943, R=0.973, F1=0.958
• Classe 8: P=0.969, R=0.984, F1=0.977
• Classe 9: P=0.487, R=0.959, F1=0.646

VALUTAZIONE QUALITATIVA:
• Performance Globale: MEDIA
• Stabilità Training: MODERATAMENTE STABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
• Considerare aumento epoche o tuning iperparametri
• Implementare class balancing o weighted loss
```